

Program: B E – Computer Science and Engineering (AI&ML) & Computer Science and Engineering (Cyber Security)
Internal Assessment – I

TERM : 03-03-2025 to 21-06-2025	COURSE NAME: No-SQL – 2
DATE : 26-04-2025 TIME: 12.30 to 01:30PM	COURSE CODE : CYAEC39/CIAEC39
MAX MARKS: 30	PORTIONS : 1,2,3 - L1-L6



Mobile Phones are banned
Instructions to Candidates:

Part A –Answer All Questions
Marks:6x1=6
Part B-Answer any two full Questions
Marks:12x2=24

Q. NO	Questions PART-A	Blooms Levels	CO	Marks
1	What is the HBase shell command to insert a value into a table? A) INSERT INTO students VALUES ('1', 'info:name', 'John') B) add 'students', '1', 'info:name', 'John' C) put 'students', '1', 'info:name', 'John' D) write 'students', '1', 'info:name', 'John'	L1	CO1	1
2	Which HBase command uses a filter to retrieve only rows where the column info:name equals "John"? A) scan 'students', {FILTER => "SingleColumnValueFilter('info', 'name', =, 'binary:John')"} B) scan 'students', {WHERE => "info:name = 'John'"} C) get 'students', {FILTER => "info:name = 'John'"} D) filter 'students', "info:name = 'John'"	L1	CO1	1
3	Which component coordinates region servers in HBase? A) Hadoop B) HDFS C) Zookeeper D) Spark	L1	CO1	1
4	Which query retrieves the top 5 highest scoring students from a Hive table? A) SELECT * FROM students ORDER BY score DESC LIMIT 5; B) SELECT TOP 5 * FROM students ORDER BY score ASC; C) SELECT * FROM students WHERE ROWNUM <= 5; D) SELECT LIMIT 5 FROM students ORDER BY score DESC;	L1	CO2	1
5	In Hive, which of the following statements is used to load data into a table? A) INSERT INTO TABLE students VALUES ('1', 'John'); B) LOAD DATA LOCAL INPATH 'students.csv' INTO TABLE students; C) IMPORT FILE 'students.csv' INTO TABLE students; D) COPY students.csv TO TABLE students;	L1	CO3	1
6	Which language does Hive use to run queries? A) Java B) SQL C) HiveQL D) Python	L1	CO3	1
PART-B				
1a	Discuss the Architecture of HBase with a neat diagram and list the application of HBase.	L1	CO1	6
b	Explain the difference between HBase and RDBMS, Describe the commonly used HBase shell command (create, scan, put and FilterList) with example.	L2	CO1	6



2a	Describe the Architecture of Hive with a neat diagram and list the features of Hive.	L3	C02	6
b	Explain the working of Hive with a neat diagram and using Hive command create an employee database with 5 field.	L2	C02	6
3a	Write an HBase query using the <i>SingleColumnValueFilter</i> to fetch rows from the Employee table where the column salary (from column family details) is greater than 50000. Discuss the role of filters in optimizing HBase scans.	L3	C01	6
b	Using the table Student (Name, USN, Course, Marks), write a Hive query to display the total number of students who scored above 75 marks. Explain how filtering (WHERE clause) works with aggregation functions like COUNT ().	L3	C03	6

Program: B E – Computer Science and Engineering (AI&ML) & Computer Science and Engineering (Cyber Security)

Internal Assessment – II

TERM : 03-03-2025 to 21-06-2025	COURSE NAME: No-SQL – 2
DATE : 18-06-2025 TIME: 03:00 to 04:00PM	COURSE CODE : CYAEC39/CIAEC39
MAX MARKS: 30	PORTIONS : 3,5,6 - L6-L14



Mobile Phones are banned
Instructions to Candidates:

Part A – Answer All Questions

Part B – Answer any two full Questions

Marks: 6x1=6

Marks: 12x2=24

Q. NO	Questions PART-A	Blooms Levels	CO	Marks
1	In HiveQL, which clause is used to filter groups after aggregation? a) where b) having c) group by d) order by	L1	CO3	1
2	Which REDIS datatype is best suited to store a collection of unique values? a) string b) list c) set d) hash	L1	CO3	1
3	What is a key feature of graph databases? a) fixed schema b) efficient storage and querying of relationships c) store data only in tabular format d) use of SQL for all queries	L1	CO4	1
4	Which neo4j clause is used to remove properties or labels from nodes? a) delete b) remove c) foreach d) limit	L1	CO4	1
5	In HiveQL, what is the main difference between JOIN and GROUP BY? a) JOIN combines rows from two or more tables; GROUP BY aggregates rows based on column values b) JOIN filters rows; GROUP BY sorts rows c) JOIN sorts rows; GROUP BY combines rows d) JOIN aggregates rows; GROUP BY filters rows	L1	CO5	1
6	Which of the following is NOT a common data model used in NoSQL databases? a) Key-value store b) Column-family store c) Document store d) Relational table store	L1	CO5	1
PART-B				
1a	Explain how GROUP BY and HAVING clauses are used in Hive. Write a query to get the departments having more than 5 students.	L2	CO3	6
b	Describe the various Redis data types. For each type, mention a use case where it would be most suitable.	L2	CO3	6
2a	Compare Relational Databases, Graph Databases, Key-Value Stores, and Column-Family Stores focusing on their schema design, query languages, use cases and performance	L4	CO4	6
2b	Explain the fundamental idea behind graph databases and highlight the key differences between graph databases and relational database systems.	L3	CO4	6
3a	Write a Cypher query using Neo4j to a node using the SET, SKIP, Aggregation clause. Explain the syntax and effect of the query.	L3	CO5	6
3b	Write Cypher queries to create 5 Category nodes (Electronics, Books, Fashion, Techs, Toys), iterate over a list to create Product nodes with names, set a price property on each product, and connect them to categories using BELONGS_TO relationships. Then, find and count products whose names contain the letter "T" and finally return all products sorted by price in ascending order.	L3	CO5	6



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(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END EXAMINATIONS - JULY / AUGUST 2025

Program : **B.E. - CSE (Cyber Security) / CSE (Artificial Intelligence and Machine Learning)** Semester : **IV**
Course Name : **NOSQL - 2** Max. Marks : **50**
Course Code : **23CYAEC49 / 23CIAEC49** Duration : **2 Hrs.**

Instructions to the Candidates:

- PART - A** : Answer all the questions.
- PART - B** : Answer one full question from each unit.

PART - A

- In HBase, what is the main use of the SCAN operation? CO1 (01)
 - To write a new row to the table.
 - To fetch a single row using its row key.
 - To iterate over rows in a table based on filter criteria.
 - To delete rows from a table
- Which filter in HBase is suitable for selecting rows where a specific column's value exceeds 100? CO1 (01)
 - FamilyFilter
 - SingleColumnValueFilter
 - ColumnPaginationFilter
 - RowFilter
- What is the correct HiveQL command to define a new table schema in Hive? CO2 (01)
 - CREATE SCHEMA TABLE
 - DEFINE TABLE
 - CREATE TABLE
 - MAKE DATABASE
- Choose the correct statement about Apache Hive: CO2 (01)
 - Hive is optimized for transactional workloads.
 - Hive allows SQL-like queries on large-scale data stored in Hadoop.
 - Hive runs independently of the Hadoop ecosystem.
 - Hive stores data in key-value formats like NoSQL systems.
- What is the result of executing the Hive query: `SELECT COUNT(*) FROM departments;` CO3 (01)
 - A list of departments with employee counts
 - The total number of department names
 - A detailed record of all departments
 - The number of rows in the departments table
- How would you calculate the average age of employees in each department using Hive? CO3 (01)
 - `SELECT AVG(age), department FROM employees;`
 - `SELECT department FROM employees WHERE AVG(age);`
 - `SELECT department, AVG(age) FROM employees GROUP BY department;`
 - `SELECT department, COUNT(age) FROM employees GROUP BY department;`

23CYAEC49/23CIAEC49

7. Which is the best definition for the graph database? CO4 (01)
A) A system that organizes data in collections and documents.
B) A system that stores information in tabular formats.
C) A database built on the concept of graph theory using nodes and edges.
D) A relational database system optimized for joins.
8. What makes Neo4j distinct as a database system? CO4 (01)
A) It uses JSON-based querying.
B) It integrates natively with MySQL.
C) It leverages graph structures and the Cypher query language.
D) It's primarily used for time-series analysis.
9. In Cypher (Neo4j), which clause is used to find patterns in the graph? CO5 (01)
A) MATCH
B) FIND
C) QUERY
D) SEARCH
10. Why would a Neo4j node have both :Customer and :VIP labels? CO5 (01)
A) To increase indexing performance by duplicating labels
B) To allow querying the node as either :Customer or :VIP
C) Because each node must have a minimum of two labels
D) To prevent the node from being matched in generic queries

PART - B

UNIT - I

1. a) Draw a clear diagram of HBase architecture and explain each major component. CO1 (04)
b) Compare HBase with relational databases by discussing the following aspects like: schema flexibility, scalability, and consistency. CO1 (04)
2. a) Suppose you want to retrieve employee data from HBase where the department (details:dept) is 'IT' and the bonus (salary:bonus) is within the 5,000 to 10,000 range. Explain how this requirement can be implemented using HBase filters. CO1 (04)
b) How would you use the HBase shell to scan the employee table and return only those records where the name column starts with the letter 'S' and the salary:basic value exceeds 50,000? CO1 (04)

UNIT - II

3. a) Explain the components and main features of Hive architecture with a diagram. CO2 (04)
b) Describe how Working of Hive, supported by a well-labeled diagram. CO2 (04)
4. a) Write a Hive query to find the total sales amount for each product in the 'Electronics' category. CO2 (04)
• products(product_id, product_name, category)
• sales(product_id, sale_date, quantity, price)
b) Write a Hive query to find the total purchase amount for each vendor in the 'North' region where the total exceeds ₹200,000. Show the vendor name, region, and total purchase amount. CO2 (04)
• vendors(vendor_id, vendor_name, region)
• purchases(purchase_id, vendor_id, purchase_date, amount)

23CYAEC49/23CIAEC49

UNIT – III

5. a) Write a Hive query to find the total number of books written by each author. Display the author name and the total count of books. CO3 (04)
- Books (book_id, title, author_id)
 - Authors (author_id, author_name)
- b) What is the difference between the WHERE and HAVING clauses in Hive? Write a Hive query to find courses which is enrolled more than 10 students. CO3 (04)
6. a) Given two tables, students(student_id, student_name, class_id, marks) and classes(class_id, class_name), write a HiveQL query to join them on class_id and select each student's name, class name, and marks. CO3 (04)
- b) What is Redis? Describe its main data types — String, List, Set, Hash, and Sorted Set — and provide an example for each. CO3 (04)

UNIT – IV

7. a) Explain the fundamental idea behind graph databases and highlight the key differences between graph databases and relational database systems. CO4 (04)
- b) Explain how you can store a simple project management graph (tasks and their dependencies) in a graph database. Write a query to find all tasks that directly depend on a given task. CO4 (04)
8. a) Write a query to create and find all Product nodes with a price greater than 100 and category as "Electronics." CO4 (04)
- b) Write a query to create and find all Employee nodes with a salary greater than 70,000 who have at least one: MANAGES relationship to a Project node name "Apollo." CO4 (04)

UNIT- V

9. a) Write Cypher queries to create 5 Person nodes with properties (user_ID, name, age, email, location), connect them with FRIEND relationships, add a city property to each, remove the city property from persons living in "Bangalore", and delete persons older than 25. CO5 (04)
- b) Write Cypher queries to create 5 Actor nodes and 5 Movie nodes, connect them with ACTED_IN relationships while setting a role property for each relationship. Find movies whose titles start with "F", count the number of movies each actor has acted in, and return the top actors ranked by their number of movies using ORDER BY, LIMIT, and SKIP. CO5 (04)
10. a) Write Cypher queries to create City nodes for the list ["Berlin", "Delhi", "Madrid", "Rome", "Osaka"] using FOREACH, assign a region property based on city name (e.g., "Asia" for "Delhi" and "Osaka"). Find and return cities starting with "D", delete cities in the "Europe" region, and finally count and return the number of remaining cities as city_count. CO5 (04)
- b) Write Cypher queries to create 5 Category nodes (Electronics, Books, Fashion, Techs, Toys), iterate over a list to create Product nodes with names, set a price property on each product, and connect them to categories using BELONGS_TO relationships. Then, find and count products whose names contain the letter "T" and finally return all products sorted by price in ascending order. CO5 (04)

SEMESTER END EXAMINATIONS – JULY / AUGUST 2025

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Course Code	: 23CYAEC49 / 23CIAEC49	Duration	: 2 Hrs.

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23CYAEC49/23CIAEC49

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